

GENERAL SPECIFICATION G2.4

Standards

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G2.4 STANDARDS

G2.4.1 Standards

(a) Provision of Relevant Standards

The Contractor shall provide copies of the relevant Standards as stipulated in the Contract for the S.O.'s reference. The Standards shall be kept in the S.O.'s Site Office and shall remain the property of the Contractor. The Contractor is deemed to have allowed for compliance with the above requirement in his pricing.

G2.4.2 Compliance with Standards and Codes

- (a) Materials, equipment and fittings shall as far as is practical, comply in all respects to the Singapore Standards (as specified by the Standards, Productivity & Innovation Board, British Standard Specification, European Standards, the standards specified herein or any comparable or equal national or international standard acceptable to the S.O.. Certificates of Origin for any materials shall be made available on demand.
- (b) Wherever in the Specification any proprietary goods or materials are specified, goods or materials of alternative manufacture but equivalent quality may be considered for approval by the S.O..
- (c) All locally manufactured electrical appliances and fittings shall be approved by SPRING and bear the SPRING Mark. Preference shall be given to such approved products. Where the products are manufactured overseas, the Contractor shall make his own arrangement for testing and approval by SPRING prior to acceptance for use.
- (d) References to standards throughout this Specification shall be deemed to refer to the latest current edition.
- (e) For any element of the structural work required to be designed by the Contractor, the latest BCA requirements with regard to the implementation of Eurocodes shall be adhered to. The Contractor's PE shall prepare and endorse the Earth Retaining Structure and Systems (ERSS) design or any structural works.
- (f) The Contractor shall use the equivalent Eurocodes Standard for any Standards which have not been updated to Eurocodes in any part of this contract specifications.

G2.4.3 Reference Standards

- (a) The Works shall be carried out in accordance with the relevant quality standards, test procedures or codes of practice, collectively referred to as Reference Standards, listed in the relevant part of the General and Particular Specifications. The Contractor shall familiarize himself fully with the requirements of such standards. If no standard is indicated then the relevant British or European Standard or, in the absence of such standard, internationally recognized standard shall apply.
- (b) British Standards are published by the British Standards Institution.

- (c) All temporary works provided by the Contractor shall comply with the codes of practices. The latest code of practice shall be adhered to:
 - (i) Code of practice for temporary electrical installation – Construction and building Sites – SS CP88-1: 2001 including amendments; and
 - (ii) Code of practice for temporary housing quarters on construction Sites – SS547:2009. Housing quarters are not allowed on TWRP. However, all temporary buildings shall be provided in accordance with the standard as a minimum.

G2.4.4 Alternative Reference Standards

- (a) Other national or international standards may be considered, provided they are, in the opinion of the S.O., no less exacting than the corresponding standard quoted in the Specification. The Contractor shall demonstrate to the S.O. that the alternative standard is suitable and equivalent to the specified standard, and shall provide proof of previous successful use. The S.O. shall decide whether the use of such alternative will be permitted as a Reference Standard. When an alternative standard is proposed, the Contractor shall supply the S.O. with two copies of the relevant standard as part of the submission of technical data for the S.O.'s approval.

G2.4.5 Applicable Issue

- (a) The applicable issue of any Reference Standard shall, unless otherwise stated in the Specification, be the issue current at the date three months preceding the date for submission of the tenders for the Contract.

G2.4.6 Abbreviations for Reference Standards

- (a) The following abbreviations where used in the specification refer to standards, codes of practice and other publications published by the organizations listed:
 - (i) ACI American Concrete Institute;
 - (ii) ANSI American National Standards Institute;
 - (iii) API American Petroleum Institute;
 - (iv) ASA American Standards Association;
 - (v) ASCE American Society of Civil Engineers;
 - (vi) ASME American Society of Mechanical Engineers;
 - (vii) ASTM American Society for Testing and Materials;
 - (viii) AWS American Welding Society;
 - (ix) AWWA American Water Works Association;
 - (x) BS British Standards Institution;
 - (xi) BS-EN British – European Standards;
 - (xii) CP British Standards Institution or Singapore Standard (Code of Practice);
 - (xiii) DIN German Industrial Standards;
 - (xiv) ISA - The Instrumentation, Systems and Automation Society;

- (xv) ISO International Organization for Standardization;
 - (xvi) JIS Japanese Industrial Standards;
 - (xvii) MS Malaysian Standards;
 - (xviii) SI International System of Units;
 - (xix) SIS Swedish Standards Commission;
 - (xx) SS Singapore Standards; and
 - (xxi) WHO World Health Organization.
- (b) Reference Standards are referred to in the text of the Specification in abbreviated form (e.g. BS 5450, SS212 or SS CP40).

END OF SECTION